

Logical volume management (LVM):

- Volumes can consist of more than one disk.
- Easy resize operation.
- Easy replacement of failing disks.
- Advanced options such as working with snapshots, which allows you to create backups even if they are open.
- Easy to add new volumes.
- Easy to add many volumes.
- Up to 256 logical volumes.

Concept to Create LVM:

- 1- Partition physical storage
- 2- Create physical volume (PV) (LVM automatically segments PVs into physical extents (PE))
- 3- Create volume group (VG) (PV can only be allocated to a single VG)
- 4- Create logical volume (LV)

```
[root@node1 ~]# pvcreate /dev/sdb1 /dev/sdc1 /dev/sdd1 (label the partition for use with LVM)
```

```
[root@node1 ~]# pvdisplay
```

```
[root@node1 ~]# pvdisplay /dev/sdb1
```

```
[root@node1 ~]# pvs
```

```
[root@node1 ~]# vgcreate VG1 /dev/sdb /dev/sdc1 /dev/sdd1
```

```
[root@node1 ~]# vgdisplay
```

```
[root@node1 ~]# vgdisplay VG1
```

```
[root@node1 ~]# vgs
```

```
[root@node1 ~]# lvcreate -n LV1 -L 2G VG1
```

```
[root@node1 ~]# lvdisplay
```

```
[root@node1 ~]# lvdisplay /dev/VG1/LV1
```

```
[root@node1 ~]# lvs
```

```
[root@node1 ~]# mkfs.xfs /dev/VG1/LV1
```

```
[root@node1 ~]# mkdir data
```

```
[root@node1 ~]# mount /dev/VG1/LV1 data
```

```
[root@node1 ~]# df -h
```

Removing a logical volume will destroy any data stored on the logical volume.

```
[root@node1 ~]# lvremove /dev/VG1/LV1 (file system must be unmounted first)
```

```
[root@node1 ~]# vgremove VG1
```

```
[root@node1 ~]# pvremove /dev/sdb1 /dev/sdc1 /dev/sdd1
```

Extending Logical Volumes (no down time):

```
[root@node1 ~]# pvcreate /dev/sde1
```

```
[root@node1 ~]# vgextend VG1 /dev/sde1
```

```
[root@node1 ~]# lvextend -L +3G /dev/VG1/LV1
```

```
[root@node1 ~]# xfs_growfs /dev/VG1/LV1 (update the file system for XFS file systems)
```

```
[root@node1 ~]# resize2fs /dev/VG1/LV1 (update the file system for other file systems)
```

Or:

```
[root@node1 ~]# lvextend -r -L +3G /dev/VG1/LV1 (extend and update in one step)
```

Shrinking a volume group:

- XFS doesn't support shrinking.

```
[root@node1 ~]# umount data
```

```
[root@node1 ~]# resize2fs /dev/VG1/LV1 100M
```

```
[root@node1 ~]# e2fsck -f /dev/VG1/LV1
```

```
[root@node1 ~]# lvreduce --size -3G /dev/VG1/LV1
```

```
[root@node1 ~]# lvreduce --size -r -3G /dev/VG1/LV1
```

```
[root@node1 ~]# vgreduce VG1 /dev/sde1 (removes sde1 from VG1)
```

```
[root@node1 ~]# mount /dev/VG1/LV1 data
```

Naming Of LVM in system:

1. /dev/vg0/lv0 as you created
2. /dev/mapper/vg0-lv0 device mapper name
What's Device Mapper ?!!!

Device mapper:

- The kernel uses the mapper to connect to storage devices such as LVM, RAID, LUCKS.

```
[root@node1 ~]# ll /dev/dm-0
```

```
[root@node1 ~]# ll /dev/mapper/VG1-LV1
```

```
[root@node1 ~]# ll /dev/VG1/LV1
```

Create a LVM snapshot:

1- Check data on LVM volume.

2- Take a snapshot.

```
[root@node1 ~]# lvcreate --size 100m --snapshot --name snap /dev/VG1/LV2
```

3- Mount sanpshoot.

4- Unmount LVM.

5- Check data.

Eng. Muhammad Adel